
Replay Accuracy Is Not Enough: An Audited Closed-Loop Evaluation Protocol for Learned Robot State Observers

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Abstract

Robot observers are usually selected by estimation error on logged trajectories, although their outputs are consumed by feedback controllers. We audit this model-selection practice for dead reckoning, an extended Kalman filter (EKF), and GRU- or selective-state-space residual observers. Six principal observers are evaluated on paired replay and closed-loop protocols across 24 conditions spanning sensing blackouts, range noise, landmark sparsity, and gyro bias. Replay position RMSE preserves broad ordering (Spearman $\rho = 0.923$) but selects a suboptimal closed-loop observer in 5/24 conditions. We then test three controller-aware offline alternatives rather than assuming that task awareness solves the problem. A local controller-sensitivity metric reduces the observed failures to 3/24, but its bootstrap interval overlaps no improvement and its worst selection regret is larger; exact command disagreement and counterfactual error replay perform worse. Matched EKF-anchor ablations overturn a confounded original conclusion: selectivity is regime-dependent, while the analytic anchor is the more consistent stabilizer. Three training initializations further reveal large variability under long blackouts. Finally, we evaluate 600 seconds of UTIAS MRCLAM physical logs and a real-noise-calibrated closed-loop replay, exposing the poor map transfer of the ordered landmark-slot representation. The resulting contribution is an open evaluation protocol: report replay and closed-loop metrics, top-1 agreement, regret, matched ablations, training-seed uncertainty, and physical-log stress tests.

1 Introduction

A state estimate is rarely the final output of a robot. It is fed to a controller, which changes the future state and observations. Yet localization and learned-filter systems are commonly selected by open-loop (OL) position or heading error on fixed logs. This assumes that lower replay error implies a better closed-loop (CL) system. The assumption is convenient, but it ignores controller sensitivity, temporal error structure, and the endogenous distribution shift created by feedback. Closely related proxy failures have been studied in model-based reinforcement learning (Lambert et al., 2020); here we isolate the problem at the state-observer layer.

We consider a differential-drive robot tracking a figure-eight from biased odometry and intermittent range-bearing measurements. Our candidate set spans classical and learned designs: dead reckoning, an EKF (Thrun et al., 2005), and recurrent residual observers with GRU or compact selective diagonal SSM cores inspired by S4 and Mamba (Gu et al., 2022; Gu and Dao, 2024). The recurrent correction is anchored either to dead reckoning or the EKF, following the broader hybrid-filter principle of retaining algorithmic structure while learning difficult components (Becker et al., 2019; Jonschkowski et al., 2018; Revach et al., 2022).

36 This work began as an architecture paper. A code and claim audit changed the contribution. The
 37 original top-1 mismatch statistic omitted GRU-EKF, and its mask/selectivity ablations changed the
 38 analytic anchor at the same time. We correct both problems and ask a broader question: *what evidence*
 39 *is needed to select a learned observer for feedback deployment?* Our contributions are:

- 40 • a paired replay/closed-loop benchmark with six deployment candidates, 24 degradation
 41 conditions, and top-1 selection regret;
- 42 • an empirical comparison of pose RMSE with exact command disagreement, local control
 43 sensitivity, and counterfactual error replay;
- 44 • matched EKF-anchor ablations and a three-training-seed robustness study;
- 45 • physical-log localization and zero-shot transfer on UTIAS MRCLAM (Leung et al., 2011),
 46 plus a closed-loop replay calibrated from its sensor residuals and observation timing.

47 The result is deliberately not a claim that one offline scalar solves deployment selection. Instead, it
 48 identifies which conclusions survive an audit and releases an evaluation checklist that makes failure
 49 visible.

50 2 Setup and evaluation protocol

51 **Robot, sensing, and controller.** The state is $s_t = (x_t, y_t, \theta_t)$ with unicycle dynamics integrated
 52 at 0.1 s. A pure-pursuit controller (Coulter, 1992) acts on the estimated pose. Odometry includes
 53 multiplicative slip, angular-rate bias, and Gaussian noise. Exteroception consists of known-map
 54 range-bearing observations, with an active landmark subset and periodic complete blackouts.

55 **Observers.** The six main candidates are DR, EKF, GRU-DR, SSM-DR, GRU-EKF, and SSM-
 56 EKF. Learned observers predict a small recurrent pose residual on an analytic anchor. Inputs contain
 57 odometry increments, ordered masked landmark tuples, an anchor encoding, and the EKF covariance
 58 diagonal where applicable. The SSM is a compact input-gated diagonal recurrence, not a full Mamba
 59 block. Models are trained on 200 domain-randomized trajectories of 200 steps.

60 **Paired protocols.** In OL replay, every observer receives identical odometry and measurements
 61 generated along an expert-controlled trajectory. In CL evaluation, the controller consumes the
 62 candidate estimate, so observer error changes actions and future inputs. Four physical axes—blackout
 63 duration, range noise, active landmark count, and gyro bias—are swept over six levels with ten
 64 evaluation seeds. The primary OL and CL metrics are position RMSE and cross-track RMSE.

65 For condition c , proxy q , and observer set $\mathcal{O}$, the selected observer and regret are

$$o_{q,c}^* = \arg \min_{o \in \mathcal{O}} q_c(o), \quad \mathcal{R}_{q,c} = J_{\text{CL},c}(o_{q,c}^*) - \min_{o \in \mathcal{O}} J_{\text{CL},c}(o). \quad (1)$$

66 This separates broad correlation from the actual deployment decision.

67 **Controller-aware proxies.** Let $e_t = \hat{s}_t \ominus s_t$ and $\pi(s, t)$ be a time-indexed controller. Exact
 68 command disagreement measures $\|D^{-1}(\pi(\hat{s}_t, t) - \pi(s_t, t))\|$. Our local control sensitivity error
 69 (LCSE) is

$$q_{\text{LCSE}} = \left[\frac{1}{T} \sum_t \|D^{-1} J_\pi(s_t, t) e_t\|_2^2 \right]^{1/2}, \quad (2)$$

70 where J_π is a finite-difference controller Jacobian and D normalizes action units. We also test
 71 counterfactual error replay: a shadow plant is propagated by the controller after injecting the
 72 candidate’s logged error trace. All are computed without rerunning sensing or the observer in closed
 73 loop.

74 3 Results: no log-only scalar replaces closing the loop

75 Table 1 confirms a nuanced result. Position RMSE is a strong coarse proxy, but it changes the
 76 top-1 observer in 20.8% of conditions. LCSE improves the observed winner count, suggesting that

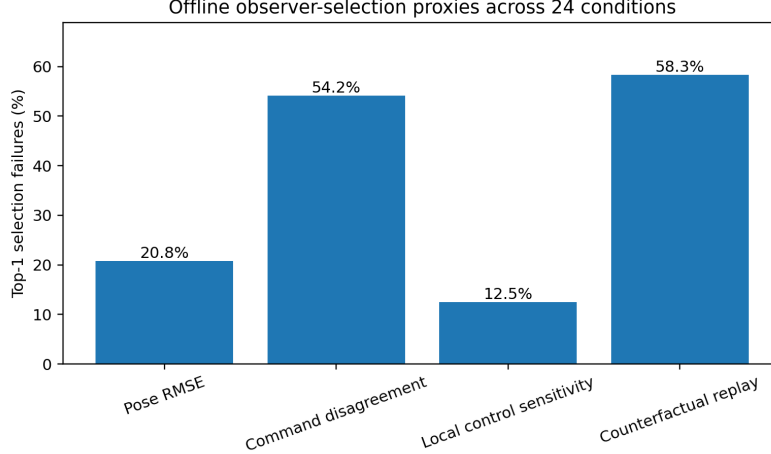


Figure 1: Observed top-1 selection failures over 24 conditions. LCSE reduces the count from 5 to 3, but a condition bootstrap gives a wide interval for the reduction; exact command disagreement and counterfactual replay are worse.

Table 1: Offline proxies against closed-loop cross-track RMSE. Regret is in meters.

Proxy	Spearman ρ	Failures	Mean regret	Max regret
Position RMSE	.923	5/24	.0051	.0347
Command disagreement	.876	13/24	.0188	.1208
LCSE	.903	3/24	.0051	.0940
Counterfactual replay	.842	14/24	.0159	.0474

77 controller anisotropy contains useful information. However, its worst regret is driven by the two-
 78 landmark condition, and a paired condition bootstrap gives a 95% interval of $[-0.125, 0.292]$ for the
 79 reduction in failure fraction. It would therefore be overclaiming to call LCSE a solved replacement.
 80 More surprisingly, exact command disagreement performs poorly: local action closeness does not
 81 capture how estimation errors modify future state and sensing. Counterfactual error replay also fails
 82 because a fixed logged error trace does not adapt to the shadow trajectory. These negative baselines
 83 are useful: adding a controller to a metric does not automatically make it deployment-valid.

84 The corrected six-observer result also retains a sharp within-model counterexample from the original
 85 study. Under compound stress, SSM-EKF reduces OL position RMSE by 43% relative to the EKF
 86 ($0.829 \rightarrow 0.472$ m) while increasing CL cross-track error by 10% ($1.324 \rightarrow 1.459$ m). GRU-EKF is
 87 best in both columns, so the defensible statement is not that SSMs are inferior, but that a large replay
 88 improvement for one observer need not transfer to control.

89 4 What actually stabilizes the learned observer?

90 The original ablation compared a selective EKF-anchored model with non-selective and mask-free
 91 models anchored to dead reckoning. That design confounded recurrence with the strength of the
 92 analytic prior. Figure 2 retrains both ablations on the same EKF anchor. Nominal cross-track RMSE
 93 is 0.199 m for selective+mask, 0.203 m without selectivity, and 0.195 m without the mask feature.
 94 Under long blackouts, the non-selective model is better (0.235 vs. 0.336 m; paired bootstrap CI for
 95 the difference $[-0.182, -0.020]$ m), while under high gyro bias it is worse (0.393 vs. 0.314 m; CI
 96 $[0.011, 0.145]$ m). Thus selectivity is a regime-dependent bias-robustness tradeoff, not a universal
 97 stability requirement. The consistent qualitative factor is the physics anchor: DR-anchored learned
 98 observers remain much less reliable in the loop.

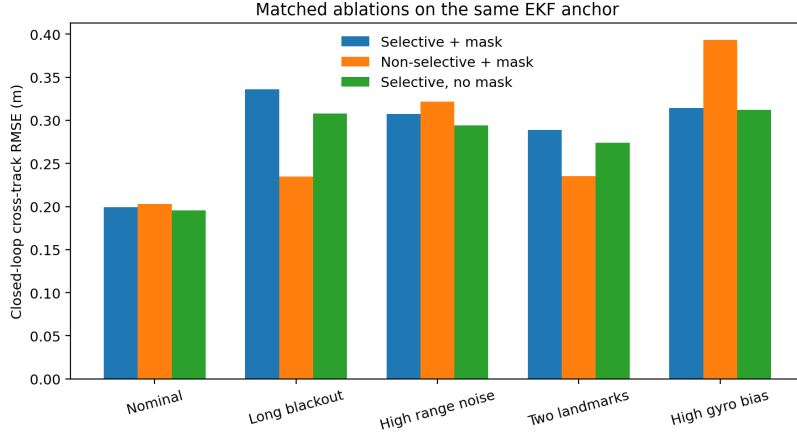


Figure 2: Matched EKF-anchor ablations. Selectivity is beneficial under high gyro bias, but non-selectivity is better under long blackouts and two-landmark sensing. Mask removal is usually close to the full model.

99 **Training-seed uncertainty.** Three SSM-EKF initializations are stable nominally (0.189–0.199
 100 m) but vary under long blackouts (0.231–0.336 m), so evaluation-seed intervals alone understate
 101 uncertainty.

102 **Controller sensitivity.** A matched five-seed Kanayama sensitivity changes the closed-loop oracle
 103 in 8/24 conditions; replay RMSE misselects 7/24 (maximum regret 0.092 m). The oracle-change
 104 bootstrap interval is wide ($[0.208, 0.625]$), so this remains secondary evidence of controller depen-
 105 dence.

106 5 Physical-log and real-noise stress tests

107 On 600 s of UTIAS MRCLAM Dataset 1 Robot 1 (6,000 aligned steps; 1,878 landmark observations),
 108 a 15-landmark EKF reaches 1.357 m position RMSE versus 3.392 m for dead reckoning (Leung et al.,
 109 2011). This is physical sensing with Vicon ground truth, not hardware closed-loop control. Zero-shot
 110 application to an eight-landmark subset exposes the ordered-slot limitation: SSM-EKF/GRU-EKF
 111 improve position modestly (2.625/2.597 m vs. 2.822 m) but worsen heading (1.65–1.66 vs. 1.51
 112 rad). A physical-residual-calibrated simulation also makes dead reckoning the oracle while controller-
 113 aware proxies incur 0.12–0.20 m regret. These are failure-oriented transfer tests, not claims of
 114 successful deployment.

115 6 Discussion and conclusion

116 We release an audited protocol: pair common-input replay with stratified closed-loop tests; report
 117 rank, top-1 agreement, and regret; use matched ablations; separate environment/training seeds; and
 118 distinguish physical logs, calibrated simulation, and hardware control. Replay RMSE changes the
 119 deployment winner in 5/24 conditions, and task-aware proxies do not consistently replace rollouts.
 120 The analytic anchor is more reliable than selectivity, while seed, controller, and MRCLAM tests
 121 expose hidden uncertainty. Known-map/association and simulated-dynamics assumptions remain; at
 122 least a small closed-loop validation is therefore required.

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